

Review retrospective:
*cumulants in moment form, and the close of the
anharmonic-arc sweep*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/OneD/AnharmonicCumulantMomentForms.lean`: the third and fourth connected cumulants of $u = -(tx)$ re-expressed in physically transparent moment form, $\kappa_3 = \langle u^3 \rangle_0 - 3\langle u^2 \rangle_0 \langle u \rangle_0 + 2\langle u \rangle_0^3$ and the analogous five-term κ_4 , with $\langle u^k \rangle_0$ the unperturbed Gibbs expectation. Result: **a clean fidelity pass, no edits**. This is the last unreviewed file of the one-dimensional anharmonic arc, so it closes a nine-file review sweep.

1 Setting

The arc proved κ_3, κ_4 first as rational closed forms in the partition values $G_n(0)$ (the form the asymptotic analysis needs), and then — in the file reviewed here — in the moment form a physicist reads off directly, $\kappa_n = (\text{raw-moment expansion of } \langle u^k \rangle_0)$. The two forms are the same object dressed differently; this file is the bridge, and the κ_2 moment form lives in its sibling `anharmonic_susceptibility_eq_connected_variance` (κ_2 review).

2 What was reviewed

Two public theorems: `anharmonic_third_cumulant_moment_form` and `anharmonic_fourth_cumulant_moment_form`, each equating `iteratedDeriv (n-1)` of the mean $M = G_1/G_0$ at 0 (the same LHS object as the rational cumulant theorems) with the raw-moment expansion in $\langle u^k \rangle_0 = \text{gibbsExp} \dots (\lambda x. (-(tx))^k)$.

3 Fidelity / soundness findings

Sound; GPT found no mismatch. The coefficients were the focus and are the standard raw-moment cumulant expansions:

- κ_3 : $m_3 - 3m_2m_1 + 2m_1^3$, coefficients $(1, -3, +2)$.
- κ_4 : $m_4 - 4m_3m_1 - 3m_2^2 + 12m_2m_1^2 - 6m_1^4$, coefficients $(1, -4, -3, +12, -6)$.

Both close the same way: the helper $\langle u^k \rangle_0 = G_k(0)/G_0(0)$ (moment-normalisation composed with `iteratedDeriv k (G_0) 0 = G_k(0)`) rewrites every Gibbs expectation to a partition ratio, the rational cumulant theorem rewrites the LHS, and `field_simp` clears the $G_0(0)$ denominators — recovering, for κ_3 , $G_3G_0^2 - 3G_2G_1G_0 + 2G_1^3$ over G_0^3 , byte-identical to the reviewed rational form, and analogously for κ_4 . The observable is written uniformly as $(-(tx))^k$ (including the $k = 1$ case as $(-(tx))^1$ rather than $-(tx)$) — a presentation choice, faithful.

4 Simplifications applied

None. GPT’s six proposals were all proof-refactoring or golf — factor the duplicated `hmom` and `hne` locals into private helpers, `simp_rw [hmom]` instead of the explicit `hmom 4 ...hmom 1` list, `norm_num→simp` on a trivial side goal, `field_simp→field_simp [hne]`, and a descriptive rename of `hmom`. None removes dead code, subsumes a hand-rolled argument with a Mathlib lemma, or drops a hypothesis; helper-extraction and golf are the declined classes. The file was left byte-identical.

Two of GPT’s *non*-findings are worth recording, because they are the right calls: `open MeasureTheory` is **load-bearing** here (the statements write `(volume : Measure ℝ)` with unqualified `Measure/volume`, thirteen times) and `hne` is **not dead** (it is the $G_0(0) \neq 0$ side condition `field_simp` consumes from context). The contrast with the immediately preceding review is sharp: there `open MeasureTheory` was genuinely dead and removed; here the same open is essential. The discriminator is, as always, whether any name or notation the open provides is actually used — and here `Measure/volume` are right there in the statements.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **Two adjacent files, same open, opposite verdict — and that is fine.** A dead-open finding is per-file, never per-name-globally: `open MeasureTheory` was dead in the cumulants-from-log-partition file (no `Measure/volume/∫` anywhere) and load-bearing here (both names in the statements). The grep is the test, and it must be run against *this* file’s text, not a habit carried from the last one.
- **Closing an arc-level sweep is itself a result.** With this file the entire one-dimensional anharmonic arc — nine files, from the moment-normalisation foundation through the FDT asymptotics, the $\kappa_2/\kappa_3/\kappa_4$ ladder, the cumulant generating function, and now the moment forms — has been independently reviewed end-to-end. Eight clean-no-edit passes and one dead-open removal; zero mathematical errors found across the arc. That uniformity is the headline, not any single file.

7 Follow-ups

- The OneD anharmonic arc review is complete. Natural next coverage: the `Multi/` multivariate files (`AffineBilinearCov`, `QuadraticApprox`) and the remaining `TwoD/` asymptotic families, none yet reviewed.